# Observable Compliance, Hidden Distortions: Cropland Protection under Incomplete Monitoring

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ASAE slides

Latest paper

# When Policy Enforcement Is Decentralized

- The **center** sets **multiple targets** for **local governments**.
- **Multi-task local bureaucrats** are **promotion-driven**: target performance shapes their **careers**.
- With **asymmetric information**, local governments may deviate.

⇒ The center **strengthens monitoring** of previously neglected tasks.

⇒ When **monitoring is incomplete**, tighter supervision may **distort** behavior rather than correct it.

**RQ**

- When the center tightens supervision on what it can observe, does it induce **within-task distortion** on what it cannot observe?

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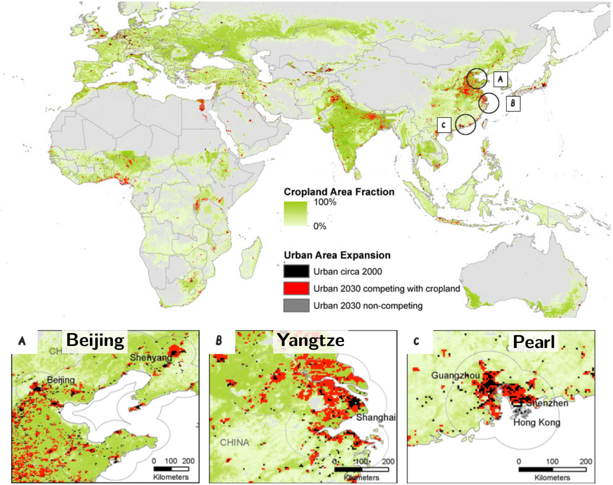
# Cropland loss from urban expansion


A
B
C
Cropland Area Fraction
100%
0%
Urban Area Expansion
⬛ Urban circa 2000
🟥 Urban 2030 competing with cropland
▩ Urban 2030 non-competing
A
Beijing
CHINA
Shenyang
Beijing
0 100 200
Kilometers
B
Yangtze
Shanghai
CHINA
0 100 200
Kilometers
C
Pearl
Guangzhou
Shenzhen
Hong Kong
0 100 200
Kilometers

Source: Adapted from Bren d'Amour et al. (2017), PNAS, Figs. 1–2. Red indicates projected urban expansion on cropland by 2030.
Panels: A = Beijing-Tianjin-Hebei; B = Yangtze River Delta; C = Pearl River Delta.

# China's Policy Response: Cropland Occupation–Compensation Balance

## Central government's objectives

- **Economic growth (GDP)**: urban expansion requires construction land.
- **Food security**: protect cropland to sustain agricultural production.

## Cropland Occupation–Compensation Balance Policy

When cropland is converted to construction land, local governments must compensate by creating an **equal amount** of cropland elsewhere.

## Incentive tension

- **Multitask bureaucrats:** GDP (urban expansion) $\leftrightarrow$ cropland protection.
- **Within-task distortion** on cropland protection: quantity (monitoring strengthened after 2017) vs. quality (monitoring not strengthened).

# Research Overview

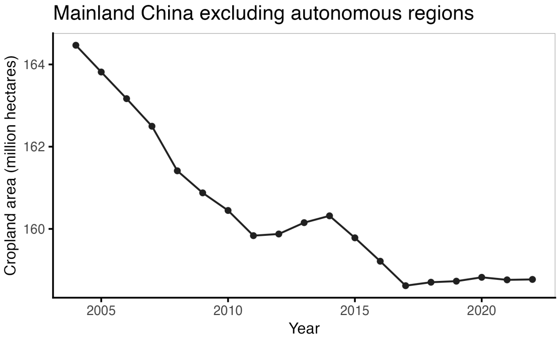
- **RQ**: When cropland monitoring becomes more stringent on *area* but not *quality*, does local implementation distort land allocation?
- **Background**
  - 2017 policy tightening: stricter enforcement of cropland area targets.
  - Jiangsu: high conflict between agricultural responsibility and economic-development pressure.
- **Data**
  - Satellite imagery + Admin data.
  - Unit of analysis: $500 \times 500\,\text{m}$ grid.
- **Method**
  - **DiD**: high- versus low-pressure counties before/after 2017.
  - **Spatial RD-DID**: nearby grid cells across administrative borders.
  - Pressure = predicted compensation burden on a county's rural land reservoir.
- **Results**
  - Quantity ↑, but new cropland is **less suitable** and **used less** (fallow ↑).
  - Quality losses go beyond exhausting good land: **distorted reclamation choices** toward **less-monitored** sites; larger losses under stronger **career incentives**.

# Related Literature

- **Principal-Agent Problem.** Holmstrom and Milgrom (1991); Baker (1992); Dixit (2002); Courty and Marschke (2004); Bandiera et al. (2021); Chen et al. (2018); He et al. (2020); Cao et al. (2026); Hong and Huang (2025); Fan et al. (2025); Fang et al. (2023); Axbard and Deng (2024); Fisman et al. (2025).
- **Decentralization and Local Incentives.** Montinola et al. (1995); Jin et al. (2005); Zhang (2006); Xu (2011); Li and Zhou (2005); Zhang et al. (2018); Wang et al. (2020); Wang and Yang (2025); Luo et al. (2025).
- **Land Regulation in China.** Cai et al. (2017); Henderson et al. (2022); Fu et al. (2021); Kong et al. (2024); Fang et al. (2026); Lu et al. (2026); Hong et al. (2025); Chen and Li (2025); Xing et al. (2026); Yu (2026).
- **Land Misallocation and Agricultural Productivity.** Restuccia and Santaeulalia-Llopis (2017); Adamopoulos and Restuccia (2014); Adamopoulos et al. (2022); Chari et al. (2021); Chen et al. (2022); Chen et al. (2023); Adamopoulos et al. (2024).
- **Contribution: First evidence on within-task distortion under incomplete monitoring**

# Background: Cropland Occupation–Compensation Balance

- Long-standing policy[a] – strengthened in 2017
- Core rule: occupy one unit, compensate one unit[b]
- Implementation:
  Upper-level government → lower-level government
  assigns **planned land use quota (cap)**
- **Planned land use quota**:
  **new construction land converted from agricultural land**
  = **cropland-compensation obligation**



Full mainland sample

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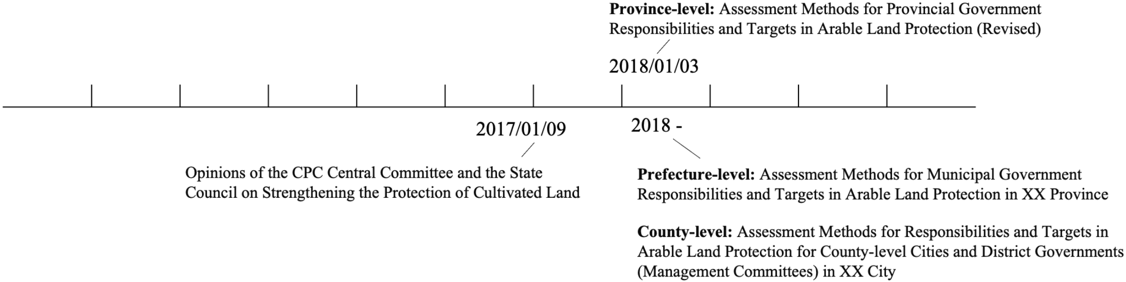
[a] (in Chinese: 耕地占补平衡政策)

[b] Compensation cropland → **permanent basic farmland**: quality shortfalls locked in.

# 2017–2018 Policy Tightening

- **Cropland Occupation–Compensation Balance**: 2017–2018 revisions assessed local cropland protection more stringently.[1]



**Province-level:** Assessment Methods for Provincial Government Responsibilities and Targets in Arable Land Protection (Revised)

2018/01/03

2017/01/09

Opinions of the CPC Central Committee and the State Council on Strengthening the Protection of Cultivated Land

2018 -

**Prefecture-level:** Assessment Methods for Municipal Government Responsibilities and Targets in Arable Land Protection in XX Province

**County-level:** Assessment Methods for Responsibilities and Targets in Arable Land Protection for County-level Cities and District Governments (Management Committees) in XX City

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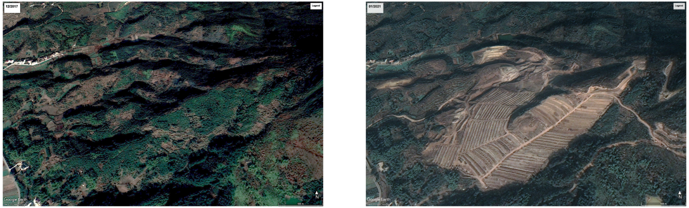
[1] Document of 2018 policy

# Background: Regulation and Local Implementation

**Implementation of the balance policy**

- **Central government's objective**: quantity-based and productive-capacity-centered.
- **Implementation distortion**: ✓ quantity, × quality.

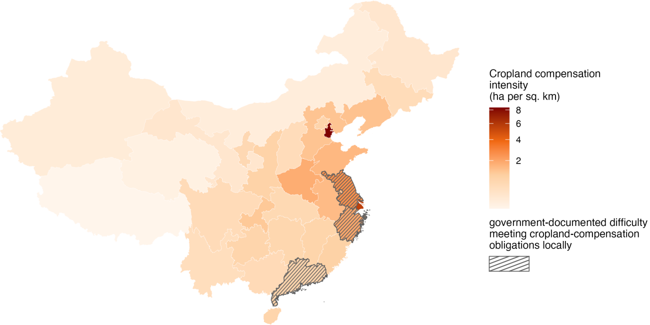


Local implementation: forest → terrace.

- **Multitask bureaucracy**: economic growth + cropland protection.
- **Land-use regulation**: government designates land for construction or agricultural use.

# Why Jiangsu?

- **Cropland-Compensation Intensity**$_p$ $=$ obligation$_p$ / provincial land area$_p$
- **Not all provinces face a binding land-use conflict.**
- **Jiangsu is a sharp case:** a major agricultural and economic province, mostly flat.


Cropland compensation
intensity
(ha per sq. km)
8
6
4
2
government-documented difficulty
meeting cropland-compensation
obligations locally

# Measurement of Key Outcomes

**Data:** satellite imagery + administrative data.

**Observation unit:** 500 m × 500 m grid cell.

1. **Cropland share:** land-cover data; fraction of 30 m pixels classified as cropland.[1]
2. **Quality / agricultural suitability:** soil-based suitability for cultivation (FAO 1976; FAO 2007).[2]
3. **Cultivation intensity:** annual Fractional Vegetation Cover (FVC).
   - *Fallow*: average annual FVC below 0.20 (bottom 25%).

Full data table

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[1] Reclaimed cropland: a grid whose cropland share is low in $t-1$ and reaches a high enough level in both $t$ and $t+1$.

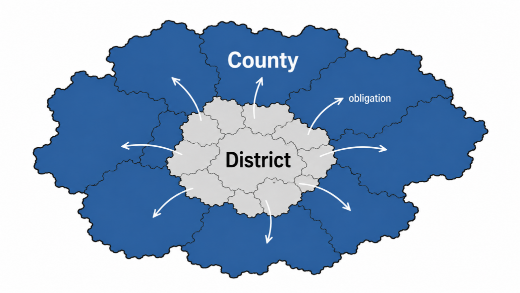
[2] The source-level indicator $1(\text{score} \geq 3)$ is 0 for saline–alkali or other non-arable land; area-weighted aggregation gives each grid's suitable-land share.

# Measurement of Cropland Protection Pressure

**Cropland Occupation–Compensation Balance** （耕地占补平衡政策）

(1) **Equal-area compensation (one-for-one).**[1]

(2) **Compensation is not necessarily local.**
  - within the same county,
  - if infeasible (e.g., urban core district), within the same prefecture (across counties)[2].



This creates a *capacity constraint* determined by the non-core (non-urban) area within a prefecture.

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[1] 占一补一: a county converting cropland to construction land compensates with an equal amount of cropland elsewhere.

[2] Jurisdictions that cannot meet the balance requirement can purchase supplementary cropland indicators through a paid, auction-based transaction platform.

# Measuring County-Level Cropland Protection Pressure

$$\text{Pressure}_c = \frac{\text{Obligation}_p}{\text{TotalLand}_p} \times \text{LandShare}_c \times 1000$$

- $\text{Obligation}_p$: planned compensation obligation of prefecture $p$'s urban core.
- $\text{TotalLand}_p$: available land outside the urban core in prefecture $p$.[1]
- $\text{LandShare}_c = \dfrac{\text{AvailableLand}_c}{\text{TotalLand}_p}$, where $\text{AvailableLand}_c$ is county $c$'s available land.

Jiangsu pressure map

Pressure distribution

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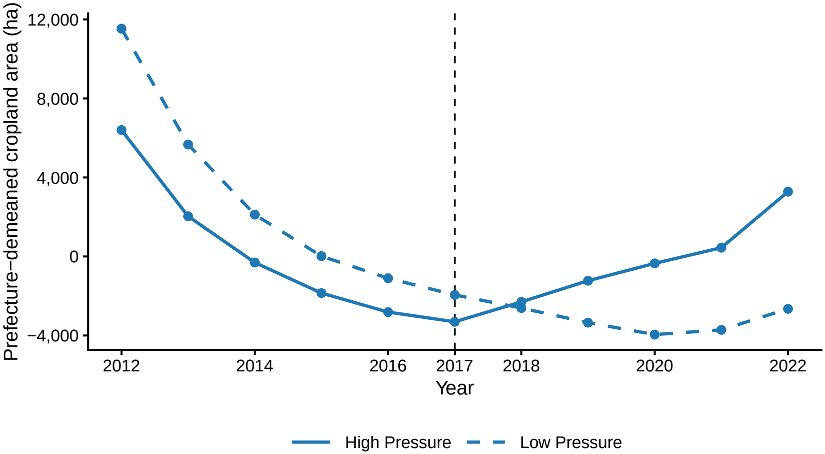
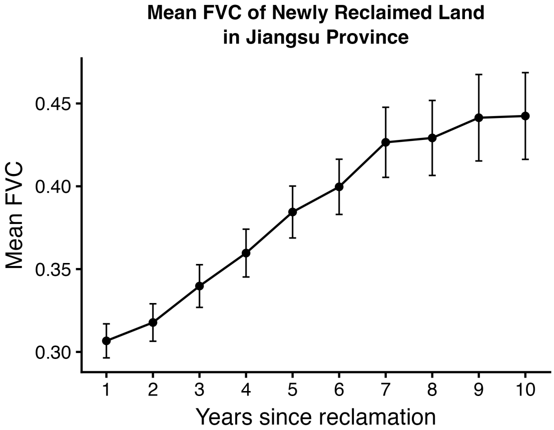
[1] Available land excludes pre-policy impervious and water shares.

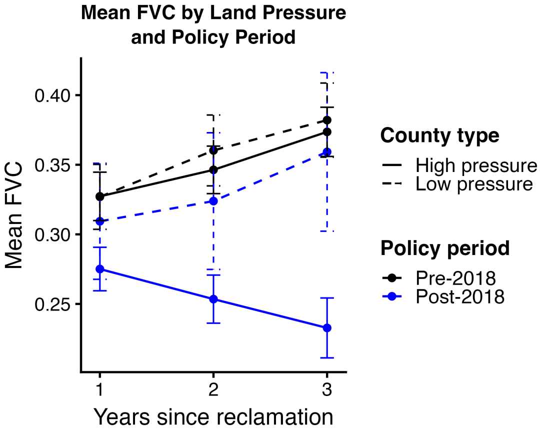
# Motivating Fact: Rural Cropland Area



# Motivating Fact: Cultivation intensity and FVC



**Mean FVC of Newly Reclaimed Land in Jiangsu Province**



**Mean FVC by Land Pressure and Policy Period**

# Identification Specification

## (1) Event-study DID: cropland share of all grids

$$\text{CropShare}_{ict} = \sum_{k \neq -1} \beta_k \big(\text{Pressure}_c \times \text{Event}_{k,t}\big) + \mu_i + \lambda_t + \varepsilon_{ict}$$

- $\text{Event}_{k,t}$: event-time indicators relative to 2018 ($k = -1$: 2017 omitted).
- **FE:** grid ($\mu_i$) + year ($\lambda_t$).

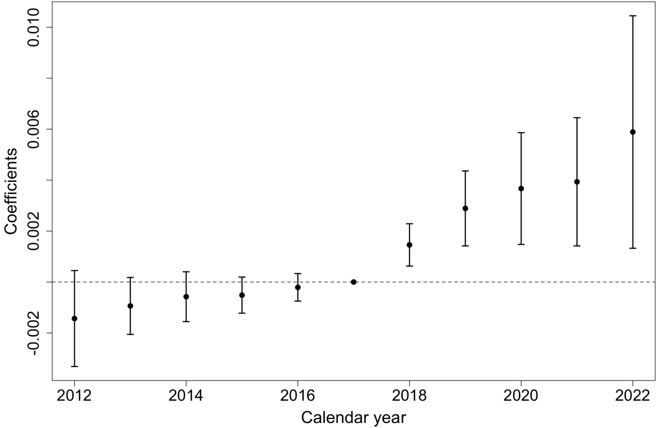
## (2) Cohort DID: fallow & soil suitability of newly reclaimed land

$$Y_{i,s} = \beta\big(\mathbf{1}\{t_0(i) \geq 2018\} \times \text{Pressure}_c\big) + \alpha_c + \mu_{t_0(i)} + \delta_s + \varepsilon_{i,s}$$

- The outcomes are attributes of the newly reclaimed land — fallow and soil suitability undefined before reclamation ⇒ cohort DID.
- **Cohort** $t_0(i)$ = grid $i$'s reclamation year.
- **FE:** county ($\alpha_c$) + cohort ($\mu_{t_0(i)}$) + years-since-reclamation ($\delta_s$).

Conceptual framework Robustness checks

# Cropland Share Expansion



- 1 s.d. $\text{Pressure}_c \uparrow \Rightarrow$ **+0.16 p.p.** cropland share in 2018 ($\approx$ **13,700 ha**); $\approx$ **84%** of the pre-period annual cropland expansion.

  Relative to 2017; pre-period annual expansion = avg. gross cropland gain 2013–2017 in the non-core grid sample ($\approx$ 16,200 ha/yr).

# Soil Suitability and Fallow Rate

| | Soil suitability | | | | 1(Fallow) | |
|---|---|---|---|---|---|---|
| | 1995 soil data | | 2023 soil data | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Pressure × Post | $-0.051^{**}$ | $-0.057^{***}$ | $-0.032^{*}$ | $-0.037^{*}$ | $0.055^{***}$ | $0.056^{***}$ |
| | (0.020) | (0.021) | (0.018) | (0.019) | (0.015) | (0.017) |
| Control Mean | 0.593 | 0.593 | 0.573 | 0.573 | 0.198 | 0.200 |
| County controls × year | | ✓ | | ✓ | | ✓ |
| County & Cohort FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Years-since-reclamation FE | | | | | ✓ | ✓ |
| Cluster level | County | County | County | County | County | County |
| Number of counties | 52 | 52 | 52 | 52 | 54 | 53 |
| Num. Obs. | 939 | 939 | 939 | 939 | 3,244 | 3,222 |
| $R^2$ | 0.452 | 0.457 | 0.490 | 0.491 | 0.232 | 0.230 |

**A 1 s.d. increase in pressure after 2017:**

- **Suitable-land share**: 9.6% lower (1995 soil data); 6.1% lower (2023 soil data).
- **Fallow rate**: 30.8% higher than baseline.

Columns (1), (3), (5); control means: 0.593, 0.573, 0.198.

Crop output per hectare Fallow event study

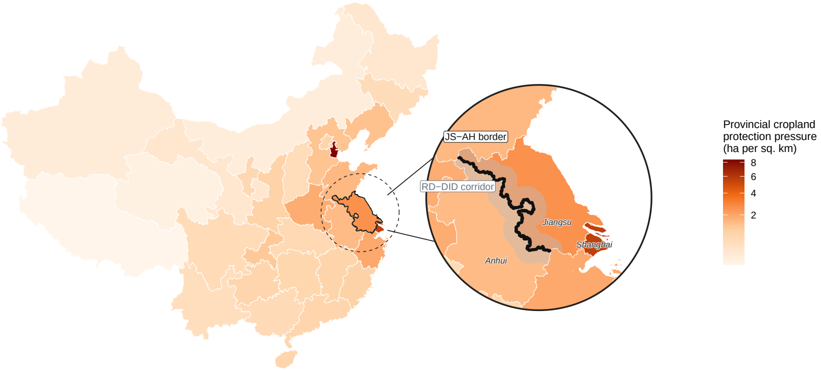
# Robustness Checks

- **Parallel trends for the cultivation-intensity measure** Event study of FVC
- **Alternative definitions of land pressure** Appendix table
- **Sensitivity to reclaimed-land classification** Appendix figures
- **Sensitivity to fallow definition** Appendix figure

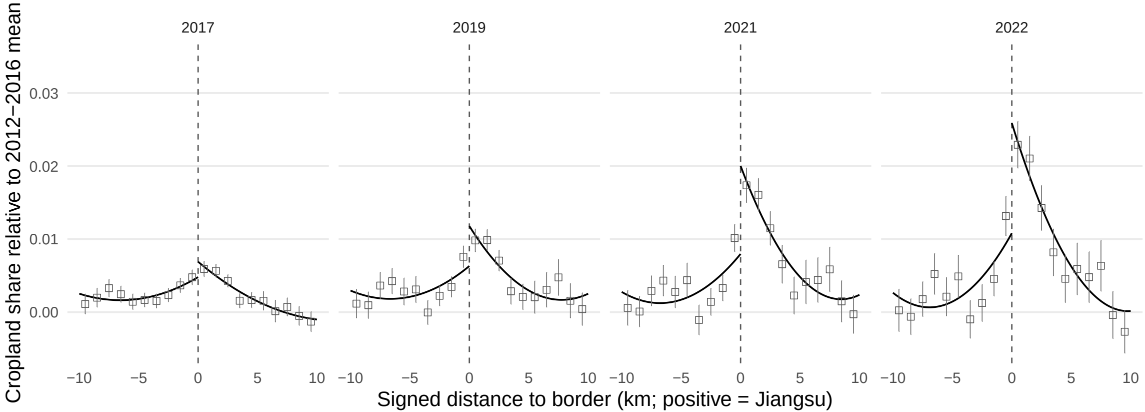
◀ Back to specification

# Spatial RD-DID

Compare pre-post change in outcome across the border.

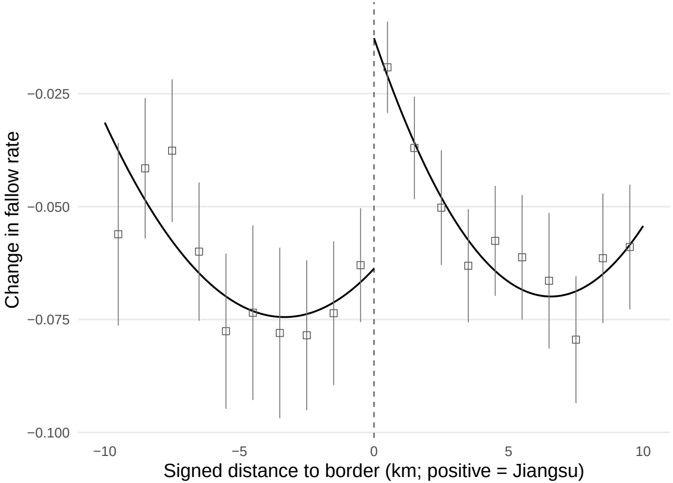

JS−AH border
RD−DID corridor
Jiangsu
Shanghai
Anhui
Provincial cropland protection pressure (ha per sq. km)
8
6
4
2

# Spatial RD-DID: Δ Cropland Share (vs. 2012–2016 Mean) at the Jiangsu–Anhui Border



Placebo: assign policy to 2013/14 ⇒ no pre-period border gap

# Spatial RD-DID: ∆ Fallow Rate at the Jiangsu–Anhui Border



# Spatial RD-DID: $\Delta$ Cropland Share and $\Delta$ Fallow Rate

| | Main comparison | | | Robustness comparison | | |
|---|---|---|---|---|---|---|
| | Jiangsu–Anhui | | | Jiangsu–Shanghai | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: $\Delta$ Cropland share* | | | | | | |
| RD in $\Delta$ cropland share | $0.006^{**}$ | $0.006^{**}$ | $0.007^{**}$ | $-0.027^{*}$ | $-0.027^{*}$ | $-0.029^{**}$ |
| (Jiangsu $-$ comparison side) | (0.003) | (0.003) | (0.003) | (0.014) | (0.015) | (0.012) |
| *Panel B: $\Delta$ Fallow rate* | | | | | | |
| RD in $\Delta$ fallow rate | $0.037^{***}$ | $0.038^{***}$ | $0.037^{***}$ | $-0.256^{***}$ | $-0.266^{***}$ | $-0.266^{***}$ |
| (Jiangsu $-$ comparison side) | (0.010) | (0.011) | (0.014) | (0.035) | (0.031) | (0.042) |
| County-pair FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Kernel | Triangle | Epanech. | Uniform | Triangle | Epanech. | Uniform |

*Notes*. Outcomes are 2018–2022 means minus 2012–2017 means. Local-linear RD-DID in pre-to-post changes within the $\pm 50$ km corridor, with additive county-pair indicators and MSE-optimal bandwidths. Entries are robust bias-corrected estimates; robust standard errors clustered by county are in parentheses. Panel B uses complete-FVC ex ante marginal cells.

# Additional Results and Alternative Explanations

**Additional results**

- New cropland is more isolated and fragmented.
- Greater conversion of water bodies and wetlands to cropland.

Full results table

**Alternative explanations for higher fallow**

- **Normal seasonality**: higher fallow persists in April–May and July–September. Evidence
- **Temporary land preparation**: no gap in years 1–2; a gap emerges in year 3. Evidence
- **Crop switching**: no clear shift among detected cultivation. Evidence

# Mechanism: Non-Distortive, Costly Compliance or Distorted Choices?

## Non-distortive, costly compliance

Quality falls because the best land has already been used.

## Distorted choices

Local governments choose less suitable land even when better options remain.

**Evidence for distorted reclamation choices**

1. **Less monitoring**: post-policy reclamation moves toward weakly monitored, far-from-station land.

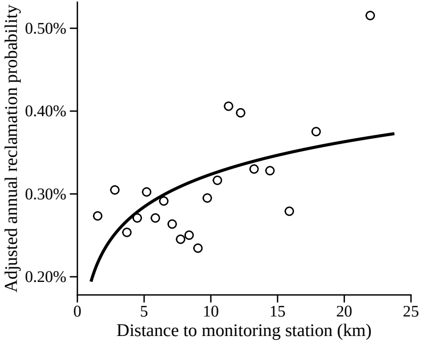
▸ Table ▸ Figure ▸ Robustness

2. **Decomposing quality losses: compliance vs. distortion**. Predict quality from historical choices, assuming unchanged selection patterns.

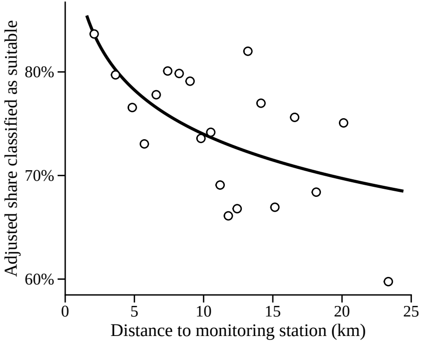
$$\underbrace{\text{Predicted quality} - \text{Actual quality}}_{\text{Distortion proxy}} = 73\text{–}78\% \times \underbrace{\text{Pre-policy quality} - \text{Actual quality}}_{\text{Total quality decline}}$$

▸ Decomposition

# Mechanism: Monitoring-Station Distance and Reclamation Choices



(a) Reclamation probability, 2018–22



(b) Suitable share of reclaimed land

- **Heterogeneity analysis** (Post $\times \log$ Distance): the shift toward parcels **farther from monitoring stations strengthens after 2017**.

▸ Table ▸ Coef. stability ◂ Back to mechanism

# Do Officials' Incentives Explain These Distortions? — Yes

| | 1995 soil data Suitable share (1) | 2023 soil data Suitable share (2) |
|---|---|---|
| Pressure × Post | $-0.097^{***}$<br>$(0.021)$<br>$[0.006]$ | $-0.067^{**}$<br>$(0.022)$<br>$[0.089]$ |
| **Incentive × Pressure × Post** | $\mathbf{-0.058}^{***}$<br>$(0.013)$<br>$[0.022]$ | $\mathbf{-0.067}^{***}$<br>$(0.021)$<br>$[0.067]$ |
| Control mean | 0.593 | 0.573 |
| Party-secretary controls | ✓ | ✓ |
| County linear trends | ✓ | ✓ |
| County & cohort FE | ✓ | ✓ |
| Number of counties | 52 | 52 |
| Observations | 939 | 939 |
| $R^2$ | 0.498 | 0.532 |
| Prefecture clusters | 13 | 13 |

- **Hypothesis**: stronger promotion incentives favor visible acreage targets over land quality.
- **Promotion incentive** $= 1$ if the prefecture party secretary is $\leq 57$ and the year precedes the 2020 national land inspection.

*Notes*. Suitable share: area-weighted source-pixel/component indicators for scores $\geq 3$. Post: after 2017. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

▸ Economic growth pressure ▸ Agricultural losses and land revenue

# Wrap-Up

- Multitasking bureaucrats generate within-task distortions under incomplete monitoring.
- The policy targets area and productivity; enforcement strengthens mainly on area.
- Satellite-based evidence from DiD and spatial RD-DID designs.
- More cropland, but lower suitability and less intensive use.
- More fragmentation and conversion of water bodies and wetlands.
- **Distorted reclamation choices**: shifts toward less-monitored sites; larger quality losses under stronger career incentives.

# Thank You!

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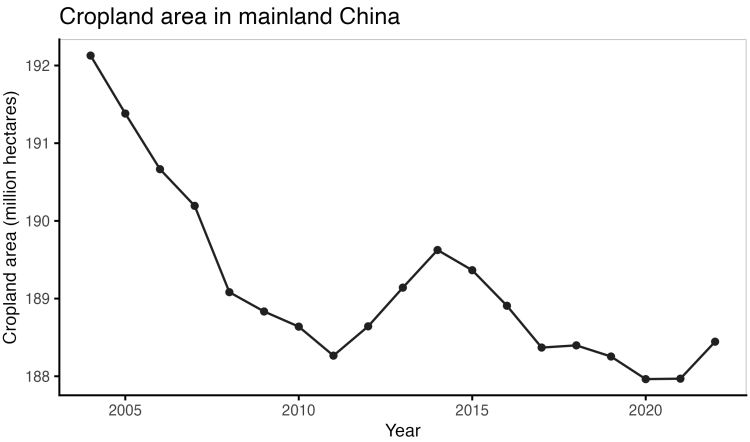
# Full Mainland Cropland Area



Data source: CLCD county data. Mainland excludes Hong Kong, Macao, and Taiwan.

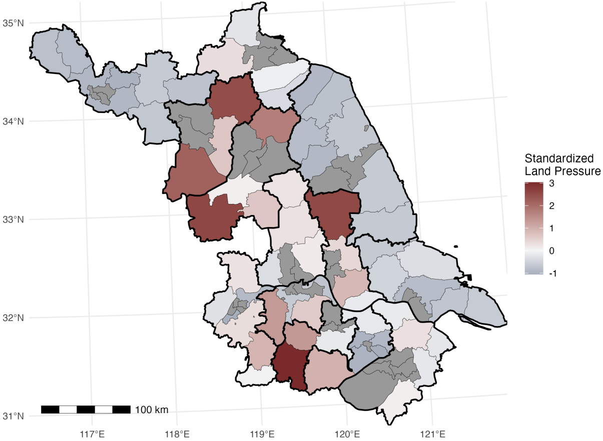
◀ Back to main figure

# Data

| Variable | Proxy | Data |
|---|---|---|
| **Remote-sensing and raster data** | | |
| Cropland | Land cover | CLCD, 30 m annual land cover |
| Cultivation intensity | Vegetation cover | FVC, 30 m / 15-day Fractional Vegetation Cover |
| Quality | Soil type | 1995 soil data and 2023 soil data |
| **Administrative and economic data** | | |
| Officials | | Public biographies of prefecture- and province-level officials |
| Boundaries / controls | | Mainland China county map; County Statistical Yearbooks |
| Planning obligations | | Prefecture Land Use Master Plans (2006–2020) |
| Land transaction | | Jiangsu Land Market transaction records |
| Growth pressure | GDP growth targets, Q1 GDP growth | Government Work Reports |

◂ Back to outcomes

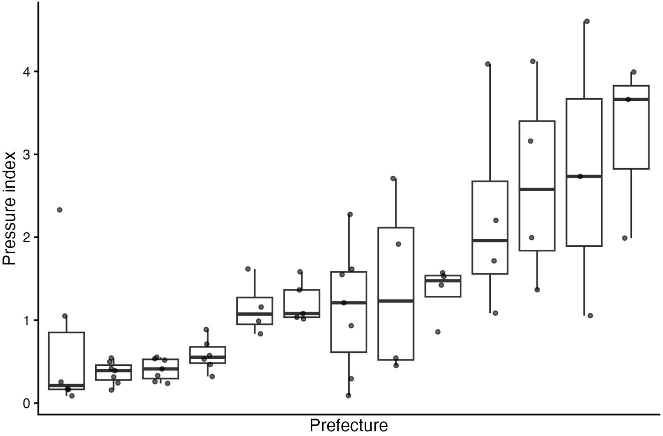
# Standardized Land Protection Pressure in Jiangsu


35°N
34°N
33°N
32°N
31°N
117°E
118°E
119°E
120°E
121°E
100 km
Standardized
Land Pressure
3
2
1
0
-1

*Note*: Positive values = above-average exposure; gray = core urban districts.

◂ Back to measurement

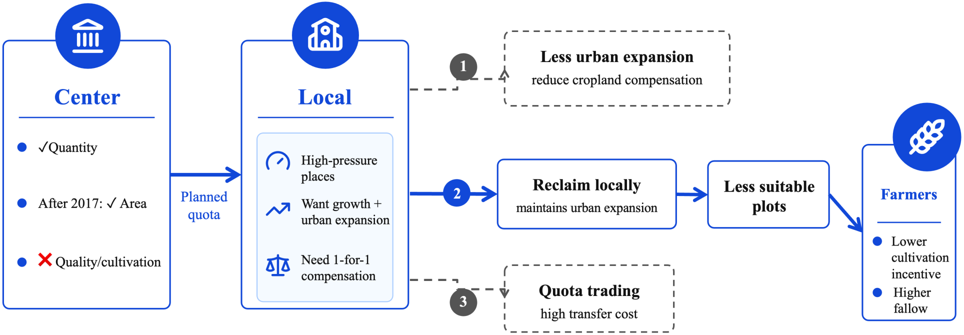
# Distribution of Cropland Protection Pressure within and across Prefectures



*Notes*. The unstandardized pressure index is shown for non-core counties. Boxes indicate the interquartile range within each prefecture, thick lines indicate medians, and dots indicate individual counties.

◂ Back to measurement

# Conceptual Framework: From Imperfect Monitoring to Misallocation



**Center**

- ✓Quantity
- After 2017: ✓ Area
- ✗ Quality/cultivation

Planned quota

**Local**

- High-pressure places
- Want growth + urban expansion
- Need 1-for-1 compensation

① **Less urban expansion**
reduce cropland compensation

② **Reclaim locally**
maintains urban expansion

**Less suitable plots**

③ **Quota trading**
high transfer cost

**Farmers**

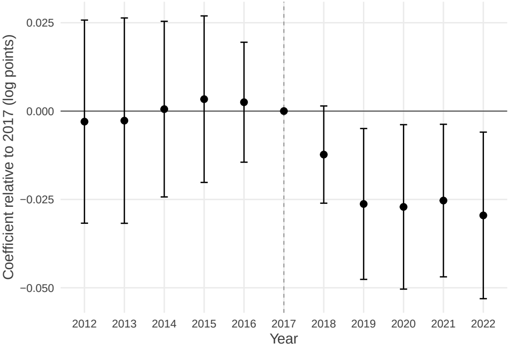
- Lower cultivation incentive
- Higher fallow

**Empirical prediction**

Higher-pressure places: ① **More compensation cropland** ② **Less suitable** ③ **Higher fallow**

◂ Back to identification

# Cultivation Intensity: Crop Output per Cropland Hectare

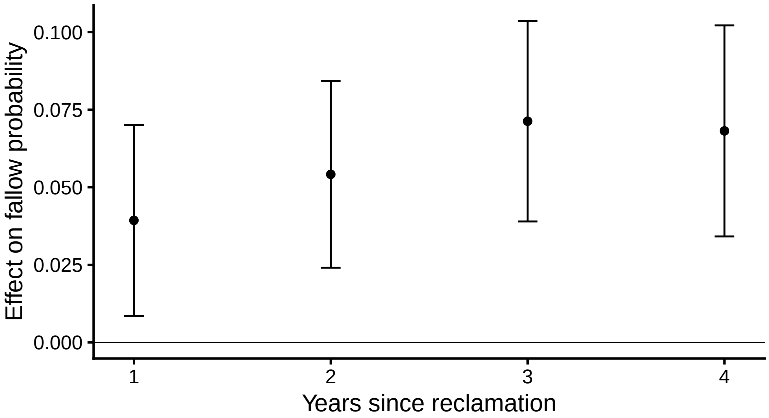


- Before 2017, crop output per hectare shows **no differential trend** $(p = 0.741)$.
- From 2019–2022, a 1 s.d. increase in pressure implies **2.5–3.0% lower output per hectare** ($p < 0.05$ each year).

County-year event study; grain + oilseed + cotton output per satellite-measured cropland hectare; county and year fixed effects; standard errors clustered by county.

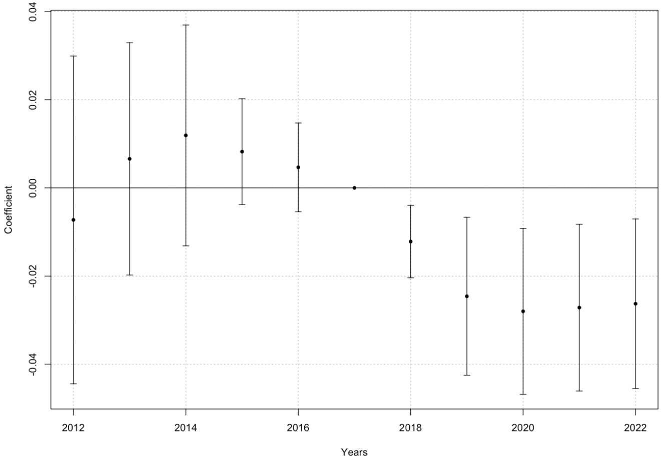
Back to outcomes

# Cultivation Intensity: Fallow Event Study



◂ Back to cultivation intensity

# Event Study of FVC



◂ Back to robustness check

# Alternative Definition of Land Pressure

Table: Robustness: Alternative Definitions of Land-Pressure Exposure

| | (1) Cropland Share | (2) Soil suitability | (3) Fallow |
|---|---|---|---|
| **Panel A: All Land Types as Available Land** | | | |
| Pressure × Post | $0.006^{***}$ | $-0.071^{***}$ | $0.078^{***}$ |
| | $(0.002)$ | $(0.024)$ | $(0.026)$ |
| Grid & Year FE | ✓ | | |
| County & Cohort FE | | ✓ | ✓ |
| Years-since-reclamation FE | | | ✓ |
| Cluster level | County | County | County |
| Number of counties | 64 | 52 | 54 |
| Num. Obs. | 3,743,982 | 939 | 3,243 |
| $R^2$ | 0.981 | 0.452 | 0.230 |
| **Panel B: Alternative Definition of Core-Urban Districts** | | | |
| Pressure × Post | $0.003^{**}$ | $-0.051^{*}$ | $0.084^{**}$ |
| | $(0.002)$ | $(0.028)$ | $(0.032)$ |
| Grid & Year FE | ✓ | | |
| County & Cohort FE | | ✓ | ✓ |
| Years-since-reclamation FE | | | ✓ |
| Cluster level | County | County | County |
| Number of counties | 64 | 52 | 54 |
| Num. Obs. | 3,743,982 | 939 | 3,243 |
| $R^2$ | 0.981 | 0.451 | 0.234 |

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

◂ Back to robustness check

# Sensitivity Analysis

- Sensitivity to Reclaimed-Land Classification
  - Baseline: $\text{Reclaimed}_t = 1$ if $\{CropShare_{t-1} < 0.60\} \wedge \{CropShare_t, CropShare_{t+1} > 0.70\}$, where $0.70$ is the pre-treatment mean.
  - Robustness: $0.60 \rightarrow 0.55$–$0.65$ for the previous year's threshold.
- Sensitivity to Fallow Definition
  - Baseline: $\text{Fallow} = 1$ if FVC $< 0.20$, where $0.20$ corresponds to the bottom $25\%$.
  - Robustness: $0.20 \rightarrow 0.13$–$0.20$.

◂ Back to robustness check

# Sensitivity to Alternative Thresholds in Reclaimed-Land Definition

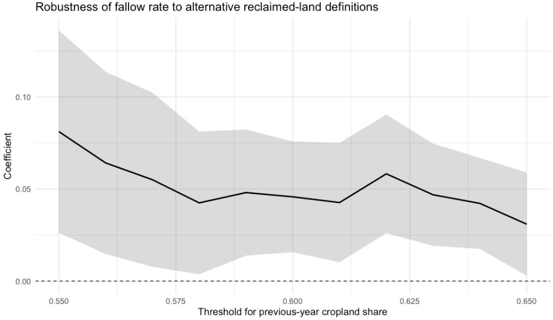


Figure: Fallow

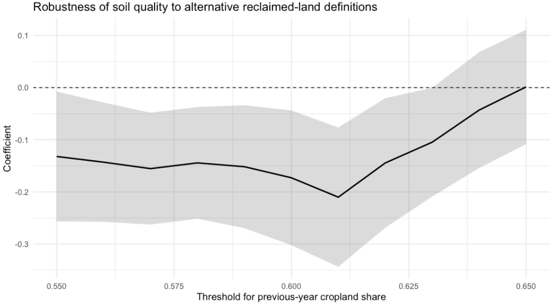
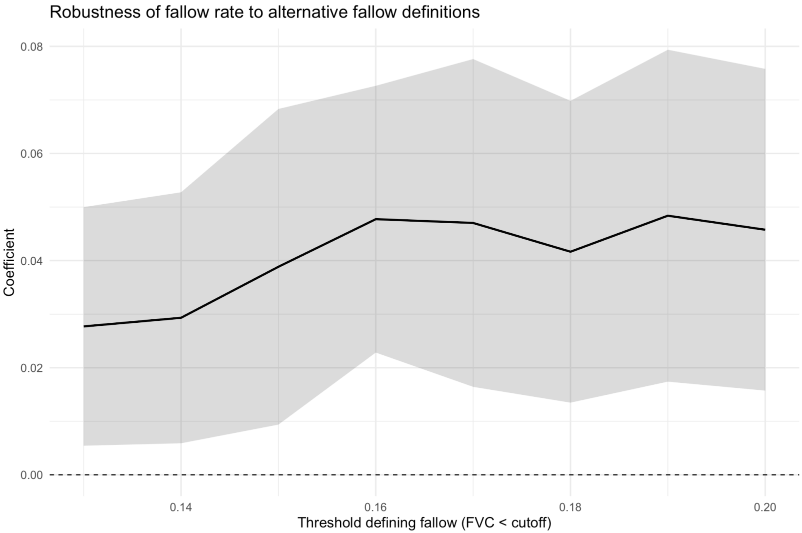


Figure: Soil Quality

◂ Back to robustness check

# Sensitivity to Alternative Fallow Rate Definition



◂ Back to robustness check

# Placebo Test

| | (1) Triangle | (2) Epanech. | (3) Uniform |
|---|---|---|---|
| *Panel A: Δ Cropland share* | | | |
| RD in 2013−2012 | −0.0016 | −0.0016 | −0.0016 |
| (Jiangsu − Anhui) | (0.0027) | (0.0026) | (0.0026) |
| RD in 2014−2012 | −0.0007 | −0.0005 | −0.0004 |
| (Jiangsu − Anhui) | (0.0034) | (0.0033) | (0.0033) |
| *Panel B: Δ Fallow rate* | | | |
| RD in 2013−2012 | 0.0112 | 0.0097 | 0.0073 |
| (Jiangsu − Anhui) | (0.0093) | (0.0100) | (0.0100) |
| RD in 2014−2012 | 0.0018 | −0.0005 | −0.0086 |
| (Jiangsu − Anhui) | (0.0157) | (0.0158) | (0.0161) |

⇒ **No pre-policy discontinuity at the border.**

*Notes*. Placebo RD-DID assigning the policy to pre-policy years (2013, 2014); the outcome is the Δ cropland share (or fallow rate) relative to 2012. No pre-policy estimate is statistically significant.

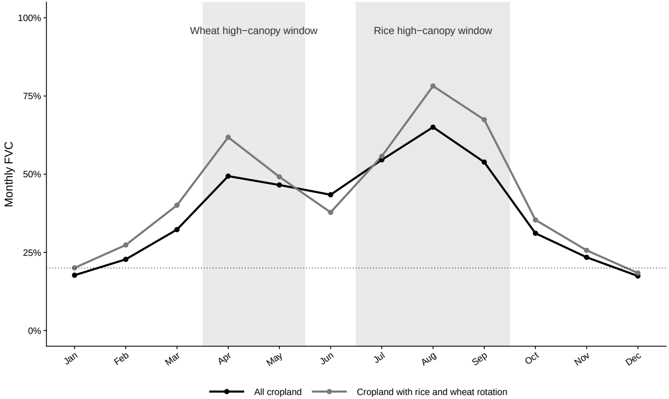
◂ Back

# Additional Results: Spatial Allocation and Environmental Outcomes

| | Spatial allocation | | | Conversion to cropland | | |
|---|---|---|---|---|---|---|
| | Dist. to existing cropland (m) (1) | Zero-cropland grids (2) | Isolated grids (3) | Water (4) | Forest (5) | Wetland (6) |
| Pressure × Post-2017 | 4.222$^{**}$ | 0.470$^{**}$ | 0.263$^{**}$ | 0.082$^{***}$ | -0.010 | 0.038$^{***}$ |
| | (1.782) | (0.184) | (0.123) | (0.018) | (0.008) | (0.009) |
| | [0.035] | [0.032] | [0.045] | $[< 0.001]$ | [0.238] | $[< 0.001]$ |
| *Magnitudes:* | | | | | | |
| 1-SD Pressure | 4.662$^{**}$ | 0.519$^{**}$ | 0.291$^{**}$ | 0.090$^{***}$ | -0.011 | 0.042$^{***}$ |
| Outcome mean | 11.011 | 1.189 | 0.311 | 0.214 | 0.046 | 0.087 |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 640 | 640 | 640 | 576 | 576 | 576 |
| $R^2$ | 0.336 | 0.324 | 0.200 | 0.690 | 0.478 | 0.587 |
| County clusters | 64 | 64 | 64 | 64 | 64 | 64 |
| Pre-policy joint $p$-value | 0.317 | 0.600 | 0.288 | 0.101 | 0.366 | 0.335 |

◂ Back to summary

# Normal Seasonality: Cultivation Calendar

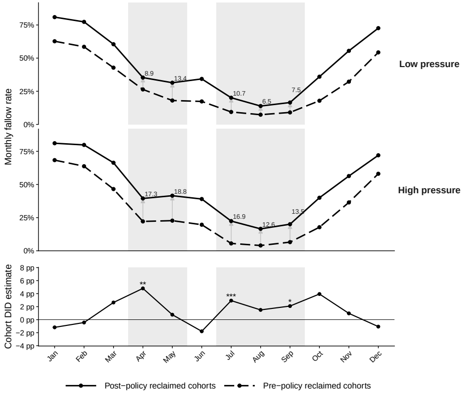


**Finding.** April–May and July–September are the high-canopy growing windows in both all cropland and rice–wheat rotation fields.

Cohort-DiD estimates

◂ Back to alternative explanations

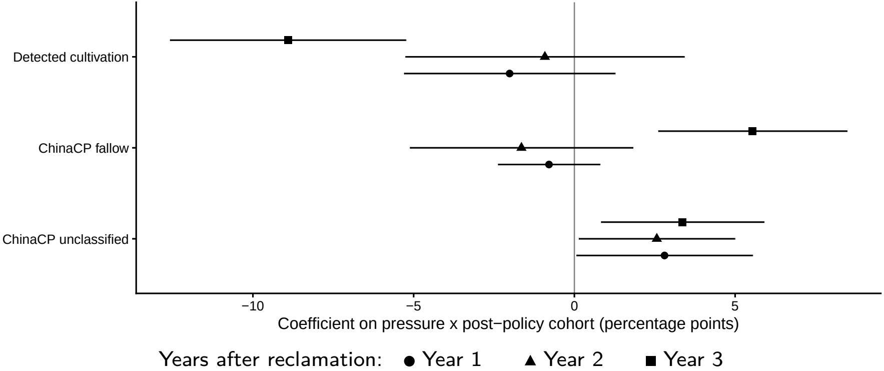
# Normal Seasonality: Month-Specific Fallow Estimates



**Finding.** In high-pressure counties, the cohort-DiD estimates are positive throughout the April–May and July–September growing windows.

Cultivation calendar
◂ Back to alternative explanations

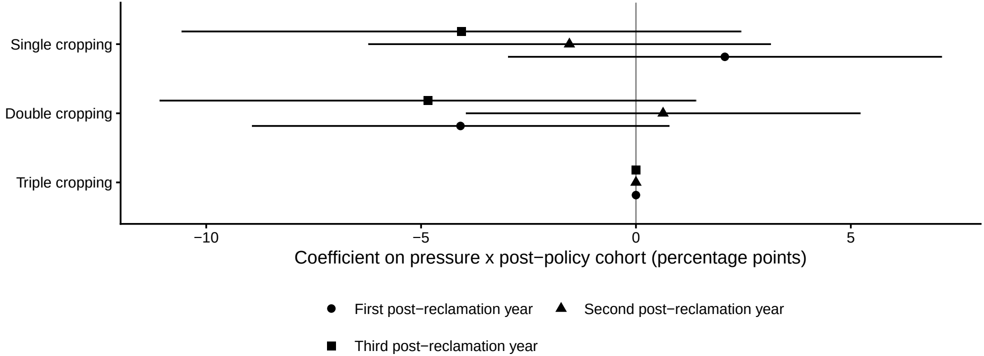
# Land Preparation: Appendix Evidence



**Finding.** If preparation drove the result, the fallow gap would peak immediately after reclamation. Instead, there is no clear difference in years 1–2; in year 3, detected cultivation falls by **8.9 p.p.** and ChinaCP fallow rises by **5.5 p.p.**

◂ Back to alternative explanations

# Crop Switching: Appendix Evidence



**Finding.** Among detected cultivation, pressure does not clearly change single-, double-, or triple-cropping shares. Nor is any crop-specific estimate significant after false-discovery-rate adjustment.

◂ Back to alternative explanations

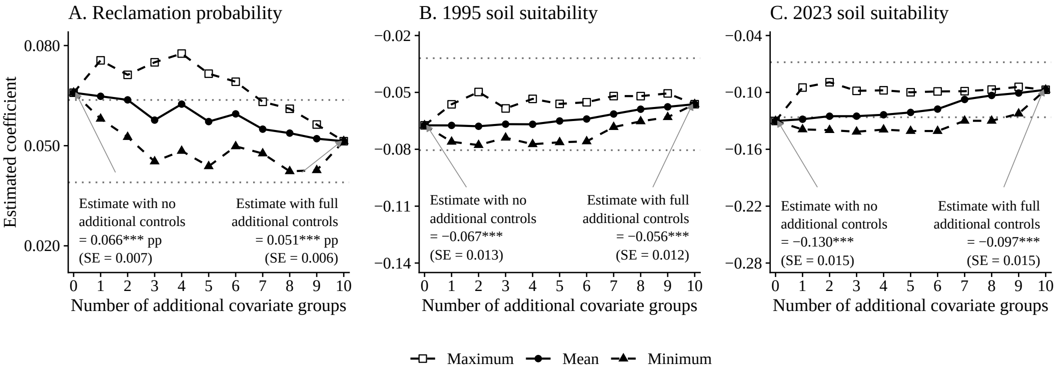
# Mechanism: Monitoring-Station Distance and Reclamation

| | Extensive margin | | | | Intensive margin: Suitable share | | | |
|---|---|---|---|---|---|---|---|---|
| | $\mathbf{1}$(Reclaimed in year) | | | | 1995 soil data | | 2023 soil data | |
| | Level gradient | | Grid FE | | | | | |
| | Baseline (1) | + Urban (2) | Baseline (3) | + Urban (4) | Baseline (5) | + Urban (6) | Baseline (7) | + Urban (8) |
| $\log(d_i) \times \text{Post}$ | $0.080^{***}$ | $0.066^{***}$ | $0.080^{***}$ | $0.066^{***}$ | $0.010$ | $0.011$ | $-0.004$ | $-0.006$ |
| | $(0.007)$ | $(0.007)$ | $(0.007)$ | $(0.007)$ | $(0.014)$ | $(0.014)$ | $(0.016)$ | $(0.016)$ |
| $\log(d_i)$ | $0.011^{**}$ | $0.014^{***}$ | | | $-0.064^{***}$ | $-0.067^{***}$ | $-0.135^{***}$ | $-0.130^{***}$ |
| | $(0.004)$ | $(0.004)$ | | | $(0.013)$ | $(0.013)$ | $(0.015)$ | $(0.015)$ |
| Grid FE | | | ✓ | ✓ | | | | |
| Year FE | ✓ | ✓ | ✓ | ✓ | | | | |
| County FE | ✓ | ✓ | | | ✓ | ✓ | ✓ | ✓ |
| Reclamation-cohort FE | | | | | ✓ | ✓ | ✓ | ✓ |
| Baseline land-use controls | ✓ | ✓ | | | ✓ | ✓ | ✓ | ✓ |
| Urban-center distance control | | ✓ | | ✓ | | ✓ | | ✓ |
| Outcome mean (share) | 0.002 | 0.002 | 0.002 | 0.002 | 0.742 | 0.742 | 0.623 | 0.623 |
| Observations | 15,512,250 | 15,512,250 | 15,512,250 | 15,512,250 | 30,891 | 30,891 | 30,892 | 30,892 |
| $R^2$ | 0.007 | 0.007 | 0.101 | 0.101 | 0.345 | 0.346 | 0.407 | 0.408 |
| Cluster level | Parent 500 m grid | | | | | | | |

*Notes.* $d_i$: distance from 100 m grid $i$ to the nearest pre-2017 provincial cropland-quality monitoring station. Cols. (1)–(4): grid-years 2013–2022, Post = 2018–2022, coefficients in p.p. Cols. (5)–(8): grids first reclaimed in 2013–2022. Soil outcomes: area-weighted shares of source pixels/components with score $\geq 3$, inherited from parent 500 m grids. SEs clustered by parent grid; ${}^{*}p < 0.10$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$.

▶ Coef. stability ◀ Back to mechanism ◀ Back to figure

# Mechanism: Coefficient Stability of Monitoring-Distance Estimates



A. Reclamation probability

Estimate with no additional controls = 0.066*** pp (SE = 0.007)

Estimate with full additional controls = 0.051*** pp (SE = 0.006)

B. 1995 soil suitability

Estimate with no additional controls = −0.067*** (SE = 0.013)

Estimate with full additional controls = −0.056*** (SE = 0.012)

C. 2023 soil suitability

Estimate with no additional controls = −0.130*** (SE = 0.015)

Estimate with full additional controls = −0.097*** (SE = 0.015)

- Add subsets of **ten covariate groups**: county-boundary and government-seat distance; soil suitability, cropland, impervious, forest, and water shares within 5 km of the station; elevation; ruggedness; nighttime lights.
- (a) $\log(d_i)\times$Post for annual reclamation (p.p.); (b)–(c) level coefficient on $\log(d_i)$ for 1995 soil data / 2023 soil data suitable shares. Lines: mean, max, min across subsets; dotted: 95% CI of the ten-group specification.
- **Signs are retained across all combinations; fully adjusted estimates remain significant.**

◀ Back to table ◀ Back to figure ◀ Back to mechanism

# Mechanism: Frontier Component vs. Selection Wedge

**Selection accounts for 73–78% of the suitability decline.**

| Reference predictors | Total loss | Frontier component | Selection wedge | Frontier share (%) | Selection share (%) |
|---|---|---|---|---|---|
| Remaining suitable share | 10.156 | 2.413 | 7.743 | 23.8 | 76.2 |
| Remaining suitable share + depletion | 10.156 | 2.204 | 7.952 | 21.7 | 78.3 |
| Remaining suitable share + depletion + reclamation intensity | 10.156 | 2.718 | 7.438 | 26.8 | 73.2 |

- **Reference**: 2002–2017 relation between each cohort's quality and the county's remaining reservoir, applied to 2018–2021 with actual quantities held fixed.
- **Frontier component**: historical mean − reference (costly compliance).
- **Selection wedge**: reference − observed quality (proxy for distortion).

*Notes*. 92 county–cohort observations in 37 counties, reclaimed-area weighted; suitable-land share (p.p.) under the 1995 soil map. Back-of-the-envelope decomposition.

◀ Back to mechanism

# Mechanism: Economic Growth Pressure

- **Growth burden**: target growth minus predicted annual growth based on quarter 1; high burden when positive and in the top tercile in Jiangsu that year.

| | Cropland share<br>(1) | New construction land area (ha)<br>(2) | Cropland-origin construction area (ha)<br>(3) |
|---|---|---|---|
| Pressure × Post × growth burden | $0.006^{*}$ | $267.707^{**}$ | $261.115^{**}$ |
| | $(0.003)$ | $(95.222)$ | $(86.557)$ |
| | $[0.358]$ | $[0.340]$ | $[0.301]$ |
| Party-secretary controls | ✓ | ✓ | ✓ |
| Grid FE | ✓ | | |
| Prefecture FE | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ |
| Observations | 3,050,309 | 116 | 116 |
| $R^2$ | 0.985 | 0.604 | 0.637 |
| Prefecture clusters | 13 | 13 | 13 |

*Notes*. Pressure: prefecture cropland protection pressure; Post: after 2017. Col. (1): non-core rural cropland; cols. (2)–(3): urban-core conveyances signed after March 31. Cropland origin uses prior-year cropland at the transaction centroid. All lower-order interactions included. Parentheses: prefecture-clustered SEs; brackets: wild-cluster bootstrap $p$-values (9,999 draws). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

◂ Officials' incentives ◂ Back to mechanism

# Agricultural Losses and Local Incentives

- A production function maps suitability and fallowing effects into agricultural value-added losses.
- A one-standard-deviation increase in pressure implies an **11.1% annual loss**.

| | Agricultural baseline | Primary-industry baseline |
|---|---|---|
| Annual loss (RMB/ha/year) | 3,456 | 5,530 |
| Capitalized perpetual loss (RMB/ha) | 59,583 | 95,346 |
| Share of conveyance revenue | 0.14% | 0.23% |

- Nearby new construction-land conveyances yield **RMB 42.1 million/ha** in gross revenue.

*Notes*. All values in 2017 RMB. Capitalization holds annual losses constant. Revenue is total deflated payments divided by area for new construction-land conveyances within 5 km of land reclaimed in the same year.

◂ Officials' incentives
◂ Back to mechanism

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